Written Homework 2: Computer Organization

CS 255: Computer Architecture and Machine Level Programming

Name: _____ NetID: _____

- **Objective**: This assignment provides guided practice reasoning about C programming concepts.
- **Submission**: Upload a completed copy of this PDF to Gradescope.
- **Grading**: Each problem is 10 points and will be graded using a check system. Showing your work is strongly encouraged.
  - Check-plus (100%): Correct solution with clear and complete reasoning.
  - Check (90%): Mostly correct, with minor error(s) or omission(s).
  - Check-minus (70%): Partially correct, with demonstrated understanding but significant mistakes or incomplete work.
  - No credit (0%): No meaningful attempt or a major conceptual misunderstanding.

1. *[Computer Overview]* Consider the following C program:

```
int x;
int y;

scanf("%d", &x);
y = x + 10;
printf("%d\n", y);
```

For each event below, identify the main component(s) involved and describe what happens to the data.

Event	**Component(s)**	**What happens to the data?**
the user enters 25		
`25` becomes the value of `x`		
`x` is obtained for the computation		
`x + 10` is computed		
the result becomes the value of `y`		
the value of `y` is displayed		

2. *[Binary Representation]* Consider the 8-bit binary string `10110110`.

   (a) Give the decimal value of the string when interpreted as an unsigned integer.

   (b) Give the decimal value of the string when interpreted as a two's-complement signed integer.

   (c) Write the string in hexadecimal.

   (d) Sign-extend the string to 16 bits.

   (e) What decimal value does your answer to part (d) represent as a 16-bit two's-complement integer? Why?

3. *[Binary Representation]* Perform each of the following operations in binary. Show your binary arithmetic, the resulting 8-bit binary value and its decimal interpretation, and state whether overflow occurs and why. Assume all values are 8-bit two's-complement signed integers.

   (a) `00110110` + `00011101`

   (b) `01100101` + `00101100`

   (c) `10110110` + `11100101`

4. *[Real Numbers]* Consider a 8-bit floating point representation with:

   - 1 sign bit,
   - 4 exponent bits (using excess-7 encoding), and
   - 3 fraction bits.

   Assume that this data type uses the conventions presented in class, including representations of zero, NaN, and (+/-)infinity.

(a) Represent the decimal value 0.8125 in this representation.

(b) What decimal value does the bit string 11010011 represent?

5. *[Real Numbers]* Answer the following questions using the same floating point representation as the previous question.

   (a) What is the largest positive representable value, other than infinity, of this representation?

   (b) What is the smallest positive representable value, other than zero, of this representation?

   (c) What is the step size (difference between adjacent representable values) between 1.0 and 2.0?

6. *[Memory]* Consider the following declarations:

```
struct Record {
    short values[3];
    int id;
    char flag;
};
struct Record records[4];
```

(a) Draw the memory layout of `records[0]`, clearly indicating padding and the end of the struct. Assume that `&records = 0x1000`.

(b) What is the starting address of `records[2]`?

(c) What is the address of `records[2].values[1]`?

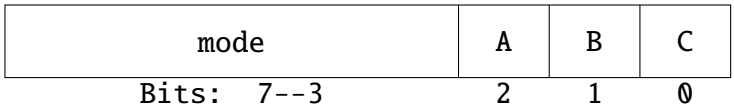
7. *[Memory]* Consider a computer system with a 16-bit address bus and a 32-bit data bus.

   (a) What is the maximum byte-addressable memory capacity?

   (b) Suppose the CPU reads a 4-byte int beginning at address 0x1008. How many memory accesses are required? What about a 4-byte int beginning at address 0x100A? Explain.

   (c) Suppose instead that the computer has a 16-bit data bus. How many memory accesses would be required to read an aligned 4-byte int? How does this compare with the original system?

8. *[CPU Structure]* An 8-bit register `status` stores several values using the following format:



The five-bit field `mode` represents an unsigned integer from 0 to 31.
Write a C expression for each of the following.

(a) Determine whether flag `B` is set.

(b) Set flag `A`.

(c) Extract the value of `mode` as an integer from 0 to 31.

9. *[Instruction Execution Cycle]* Consider the following instruction which shifts the value in `R2` left by 2 bits and stores the result in `R1`.

```
shift R1, R2, 2
```

Describe what happens during each stage of the instruction execution cycle:

(a) Fetch

(b) Decode

(c) Execute

(d) Memory

(e) Writeback

10. *[Instruction Execution Cycle]* A program executes 600 million instructions on a 3 GHz processor with an average CPI of 2.0.

    (a) How long does the program take to execute?

    (b) A processor improvement reduces the average CPI to 1.0, but requires the clock rate to decrease to 2 GHz. How long does the program take to execute on the improved processor?

    (c) Calculate the speedup from the improvement.

    (d) Explain why reducing CPI does not necessarily reduce execution time.

11. *[Caches]* A designer is considering a cache hierarchy with an L1 cache with a 25% miss rate and 1 cycle access time backed by a larger L2 cache with a 7% miss rate and 5 cycle access time. The L2 cache is backed by memory with a 100 cycle access time.

    (a) What is the average access time, in cycles, of the L2 cache?

    (b) What is the average access time, in cycles, of the L1 cache?

    (c) Suppose the L2 cache is removed, so the L1 cache is backed directly by main memory. How much slower would the average memory access time be?

12. *[Caches]* Consider a byte-addressable system with 12-bit addresses and a 64-byte direct-mapped cache. Each cache block holds 8 bytes.

    (a) How many blocks can the cache hold?

    (b) How many bits of a memory address are used for the byte offset, set index, and tag?

    (c) Determine the tag, set index, and outcome for each memory access. Assume the cache is initially empty.

Address	Tag	Set Index	Hit/Miss
0 0 0 0 0 0 0 0 0 0 0 0			
0 0 0 0 0 0 0 1 1 0 0 0			
0 0 0 0 0 0 0 0 0 0 1 1			
0 0 0 0 0 1 0 0 0 0 0 0			
0 0 0 0 0 0 0 0 0 0 0 0			
0 0 0 0 0 0 0 1 1 0 0 0			